Thomas Nguyen

(737) 245-4267 | thomasn8255@gmail.com | LinkedIn | GitHub | thomastnguyen.dev | Security Clearence (Secret)

Education

University of California - San Diego Sept 2026 - June 2028
M.S. in Computer Science and Engineering

University of California - Irvine Graduated: June 2026
B.S. Computer Science *GPA: 3.71*

Work Experience

RTX (Raytheon) June 2026 – Aug 2026
Software Engineer Intern *El Segundo, CA*

- Developed a Python-based LLM test-generation tool averaging 350 unit tests and 90% line coverage across 5 repositories.
- Built deterministic test discovery mapping 40+ source files to existing unit tests, preventing redundant test generation.
- Implemented a source-hashed cache that regenerates on file changes, cutting redundant LLM calls 50% across repeated runs.
- Engineered orchestration logic advancing the model through a state machine across discovery, generation, and validation.

UCI Design and Partnership Lab (DapLab) Oct 2025 – May 2026
Project Team Lead *Irvine, CA*

- Deployed a Dockerized Node.js backend on AWS EC2 with secure auth and data pipelines for a platform of 1000+ students.
- Automated AI prompt workflows via Node.js/PostgreSQL, using embedding-similarity checks to cut duplicate content 30%.
- Built TypeScript REST APIs with schema validation for progress persistence, enabling session restore across user visits.

SportsStake June 2025 – Sept 2025
Software Engineer Intern *Los Angeles, CA*

- Served live odds and account data through Node.js/PostgreSQL endpoints on a live sports betting app with 300+ users.
- Cut redundant PostgreSQL queries through Redis caching and cursor-based pagination, reducing latency by 30%.
- Added async error boundaries and TypeScript input validation to reject malformed requests before downstream services.
- Set up GitHub Actions CI/CD with automated test suites, catching regressions pre-merge and enabling frequent deploys.

UCI CALIT2 Oct 2024 – May 2025
Machine Learning Research Assistant *Irvine, CA*

- Built a time-series model in Python using PyTorch with windowed data loaders and batching logic for FEM simulations.
- Developed an ETL pipeline with pandas and scikit-learn, processing 600+ CSVs and cutting training time 75%, and added evaluation and visualization tooling to monitor prediction quality and compare model outputs.

Projects

PC Diagnostic Monitor | Python, C, IOKit, CrewAI, pytest, PyInstaller GitHub Repo

- Built a real-time PC monitor across 3 OSes unifying psutil, LibreHardwareMonitor, and SMC behind one `read()` interface.
- Supported 74 macOS thermal sensor identifiers by developing a native C helper with automatic HID/SMC fallback.
- Enabled concurrent dashboard reads by designing a thread-safe rolling cache retaining 300 telemetry snapshots.
- Reduced alert noise with a 5-second debounce and hysteresis; packaged standalone binaries using PyInstaller.

Dependency Graph Auditor | Go, PostgreSQL GitHub Repo

- Built a concurrent dependency auditor that crawls npm package graphs to detect and persist transitive dependencies at scale.
- Engineered a bounded goroutine worker pool with backoff-and-jitter retries, a dead-letter queue, and at-least-once delivery.
- Deduplicated cyclic and diamond dependencies in PostgreSQL while applying backpressure to rate-limited registries.
- Added graceful shutdown with drain and timeout, allowing in-flight jobs to finish safely on SIGTERM.

Technical Skills

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Design, Computer Architecture, Operating Systems, Computer Networks, Probability & Statistics, Discrete Mathematics, Web Development
Languages: Go, C++, Python, Java, JavaScript, TypeScript, SQL, Bash, HTML/CSS
Technologies: Node.js, Express, React, REST APIs, PostgreSQL, Redis, Supabase, Prisma ORM, Docker, AWS EC2, GitHub Actions (CI/CD), Git, Linux